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REVIEW

Comparative efficacy and safety of photobiomodulation, transcranial direct current stimulation, and repetitive transcranial magnetic stimulation in Alzheimer's disease: a network meta-analysis of randomized controlled trials

Wei Xu[1] · Jiajie Chen[1] · Yingying Zhou[1] · Chengzhi Liu[2] · Yuelin Han[1] · Kai Zheng[1]

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Abstract

Alzheimer's disease (AD) is a progressive neurodegenerative disorder for which current pharmacological therapies provide only modest and transient benefits, underscoring the need for effective non-pharmacological interventions. Non-invasive brain stimulation (NIBS), including photobiomodulation (PBM), transcranial direct current stimulation (tDCS), and repetitive transcranial magnetic stimulation (rTMS), is a promising approach. This study compared the efficacy, feasibility, and safety of PBM, tDCS, and rTMS in AD. We conducted a Bayesian network meta-analysis (NMA) of randomized controlled trials (RCTs), prospectively registered in PROSPERO (CRD420251006868) and reported according to PRISMA-NMA. The primary outcome was global cognitive performance, assessed by the Mini-Mental State Examination (MMSE) and the Alzheimer's Disease Assessment Scale–Cognitive Subscale (ADAS-Cog). Secondary outcomes were feasibility and safety, evaluated by dropout rates and adverse events, respectively. A total of 34 studies, comprising 35 RCTs (n = 1,559), were included, with one study contributing two independent randomized controlled trials. All three interventions improved global cognitive performance compared with control. PBM showed the highest estimated effects on both MMSE (MD = 3.14, 95% CrI: 1.71 to 4.60) and ADAS-Cog (MD = −8.01, 95% CrI: −12.82 to −3.13). In indirect comparisons, PBM was superior to rTMS for ADAS-Cog (MD = −5.24, 95% CrI: −10.23 to −0.12), while the MMSE comparison narrowly missed statistical significance (MD = 1.57, 95% CrI: −0.02 to 3.18). These findings were generally consistent across subgroups stratified by baseline cognition and intervention duration. PBM tended to rank highest for cognitive improvement with feasibility and safety comparable to tDCS and rTMS in AD. These findings support the therapeutic potential of PBM, and large-scale RCTs are needed to guide individualized treatment strategies, with implications for clinical decision-making in geriatric care.

Keywords Photobiomodulation · Non-Invasive Brain Stimulation · Alzheimer's disease · Cognition · Safety and Tolerability

✉ Kai Zheng
diazna2002@sina.com

Wei Xu
403980849@qq.com

Jiajie Chen
1005843466@qq.com

Yingying Zhou
zyy15560125332@163.com

Chengzhi Liu
zeroemia@hust.edu.cn

Yuelin Han
hanyuelin43@gmail.com

[1] Tongji Medical College, Tongji Hospital, Huazhong University of Science and Technology, 1095 Jiefang Avenue, Wuhan, China

[2] Tongji Medical College First Clinical College, Huazhong University of Science and Technology, 1095 Jiefang Avenue, Wuhan, China

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Abbreviations

AD	Alzheimer's disease
NIBS	Non-invasive brain stimulation
PBM	Photobiomodulation
tDCS	Transcranial direct current stimulation
rTMS	Repetitive transcranial magnetic stimulation
NMA	Network meta-analysis
RCTs	Randomized controlled trials
SUCRA	Surface under the cumulative ranking curve
MMSE	Mini-Mental State Examination
ADAS-Cog	Alzheimer's Disease Assessment Scale–Cognitive Subscale
LTP	Long-term potentiation
MDs	Mean differences
CrI	Credible interval
RRs	Risk ratios

Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder and the most common cause of dementia in the aging population [1]. Globally, over 35 million people are currently living with AD, and this number is projected to triple by 2050 as the population ages [2]. The pathophysiology of AD involves β-amyloid burden, tau neurofibrillary tangles, chronic neuroinflammation, and mitochondrial abnormalities [3]. Clinically, AD is characterized by progressive decline in memory, attention, language, executive function, and eventually global cognition, resulting in substantial burdens for both patients and caregivers. Current pharmacological treatments provide only modest symptomatic relief and are often constrained by poor tolerability, economic burden, and uncertain long-term efficacy [4]. These challenges are especially pronounced in older adults with multimorbidity or functional frailty. Consequently, growing attention has been directed toward non-pharmacological interventions that provide safer and more effective options for cognitive function in AD.

Among these approaches, non-invasive brain stimulation (NIBS) has gained increasing interest as a non-pharmacological alternative for cognitive improvement in AD [5]. The two most widely studied modalities of NIBS are transcranial direct current stimulation (tDCS), which modulates cortical excitability by low-intensity direct currents, and repetitive transcranial magnetic stimulation (rTMS), which induces long-term potentiation (LTP)-like synaptic plasticity through time-varying magnetic fields [6–8]. Both tDCS and rTMS have demonstrated cognitive benefits in individuals with AD, particularly in improving attention, working memory, and executive function [9]. Photobiomodulation (PBM) represents a novel NIBS technique that uses red or near-infrared light (600–1100 nm) to penetrate cranial tissues to stimulate mitochondrial cytochrome c oxidase. This photonic stimulation enhances mitochondrial ATP synthesis, reduces oxidative stress, and activates neuroprotective pathways [10]. In animal models of AD, PBM has been shown to reduce β-amyloid deposition, increase synaptic density, and improve cognitive deficits [11].

Although several meta-analyses have evaluated NIBS in AD, many have grouped distinct modalities into a single treatment category, limiting modality-specific insights [12]. Notably, a recent NMA compared tDCS and rTMS in AD, while PBM was not evaluated, leaving its relative efficacy and clinical application uncertain [13]. To date, the comparative efficacy, feasibility, and safety of PBM, tDCS, and rTMS in AD have not been systematically assessed. These three modalities are among the most commonly investigated NIBS approaches in AD. To overcome these limitations, we conducted a Bayesian network meta-analysis (NMA) to compare the efficacy, feasibility, and safety of PBM, tDCS, and rTMS in individuals with AD [14]. The findings aim to provide clearer clinical evidence and guide future research on non-pharmacological strategies for AD.

Methods

Study design and objectives

This NMA was conducted in accordance with the PRISMA extension statement for NMA [15] (Table S1). The protocol outlining eligibility, outcomes, and the analytic framework was registered in PROSPERO (CRD420251006868). Ethical approval was not required, as only published aggregate data were analyzed. This study aimed to compare PBM, rTMS, and tDCS on cognitive outcomes in AD or probable AD. Other neuromodulation modalities, including transcranial alternating current stimulation (tACS), were outside the prespecified scope of this registered review and were therefore not eligible for inclusion. The analytic framework followed the PICO model, as follows: (1) Population: Older individuals with AD/probable AD (non-AD dementias excluded); (2) Intervention: Treatment involving PBM, rTMS, or tDCS, delivered over at least one complete cycle (≥ 5 days); (3) Comparison: sham stimulation or usual care; (4) Outcome: Change in global cognitive performance, assessed by either Mini-Mental State Examination (MMSE) or Alzheimer's Disease Assessment Scale–Cognitive Subscale (ADAS-Cog) [9, 13, 16–18].

Search strategies and eligibility criteria

A systematic literature search was conducted across Cochrane CENTRAL, EMBASE, PubMed, and Web of Science to identify RCTs published from database inception to 30 June 2025 (Table S2). In order to capture relevant grey literature and unpublished studies, a search was also conducted on ClinicalTrials.gov. No restrictions were imposed on language. Furthermore, the search strategy was supplemented by citation searching (snowballing), which entailed screening the reference lists of included articles, relevant review papers, and prior meta-analyses to identify further eligible studies. Records identified through database searches and citation searching were merged before deduplication and subsequent screening. The screening process was conducted in two stages by two independent authors. In the first stage, titles and abstracts were screened to assess potential relevance. In the second stage, full-text reports were sought for retrieval and evaluated against the predefined eligibility criteria, which were as follows: (1) RCTs; (2) interventions involving at least one of the three prespecified NIBS modalities: PBM, rTMS, or tDCS; (3) participants diagnosed with AD or probable AD; (4) studies that used validated cognitive assessment tools to evaluate the effects of different NIBS interventions on cognitive impairment. Studies were excluded if they: (1) were not RCTs; (2) were conducted on non-human subjects; (3) involved participants with clearly diagnosed non-AD dementias, such as vascular dementia or frontotemporal dementia; (4) did not report quantifiable results of cognitive function assessments; (5) administered NIBS for fewer than five consecutive days or failed to complete a defined treatment course. For reports using broader terms such as dementia, eligibility was determined based on whether the article, trial registration, or participant description supported AD or probable AD. Studies explicitly enrolling non-AD dementias were excluded. When multiple reports described the same or overlapping RCT, only the more complete report was retained.

Data extraction and quality assessment

Two authors independently extracted relevant data, including study characteristics (first author, year, country, study design); participant demographics (sample size, sex distribution, age); diagnostic criteria, intervention details (modality, parameters, duration); comparators; and outcomes. To improve the transparency and clinical comparability of the included evidence, we extracted the diagnostic criteria or definitions used in each study and summarized these data in Table S3. The quality of included studies was assessed by the Cochrane RoB 2 tool [19]. Each trial was classified into low risk, high risk, or having "some concerns". Any discrepancies in data extraction or risk-of-bias assessment were resolved by consensus or, when necessary, through discussion or consultation with a third author.

Statistical analysis

The primary outcomes were MMSE and ADAS-Cog, and the secondary outcomes were dropout rate and adverse events. MMSE and ADAS-Cog were selected because they were the most frequently reported and most comparable global cognitive measures across the included trials. Studies reporting other validated cognitive outcomes were not excluded during screening; however, outcomes without extractable MMSE or ADAS-Cog were not entered into the primary pooled efficacy analysis. For all outcomes, data were extracted from the first post-intervention assessment. All available evidence regarding the efficacy and safety of NIBS interventions for AD was synthesized using mean differences (MDs) with 95% credible intervals (CrI) for continuous outcomes (MMSE, ADAS-Cog), and risk ratios (RRs) with 95% CrI for binary outcomes (dropout rates and adverse events).

Network plots were generated in R software (version 4.4.3) to illustrate the comparisons among PBM, rTMS, tDCS, and a control group (CON). The control group comprised sham stimulation and usual care [20]. Eligible multi-arm trials were handled using multi-arm models in the network analysis, with the shared control group appropriately accounted for. A Bayesian NMA was conducted using Markov Chain Monte Carlo (MCMC) methods implemented in R software (version 4.4.3) [20]. For continuous outcomes, analyses were conducted using the gemtc package (version 1.0.2) [21, 22]. A random-effects consistency model was constructed with a normal likelihood and an identity link function. Four Markov Chain Monte Carlo (MCMC) chains were run, with 10,000 iterations discarded as burn-in and 30,000 iterations used for posterior inference. For binary outcomes, analyses were performed using the BUGSnet package (version 1.1.3) [23]. A random-effects consistency model was constructed with a binomial likelihood and a log link function, employing the same MCMC configuration. Convergence was assessed using the Gelman–Rubin diagnostic, with a potential scale reduction factor (PSRF) close to 1 considered indicative of adequate convergence [24]. To validate and complement the results from the Bayesian NMA, pairwise meta-analyses (PWMA) were conducted using the meta package (version 8.1.0) in R software (version 4.4.3) [25]. The Bayesian framework also provided treatment rankings based on posterior rank probabilities. These rankings were visualized and summarized by calculating the surface under the cumulative ranking curve

(SUCRA), with higher SUCRA values indicating greater efficacy or lower risk, depending on the outcome [26].

The validity of the NMA is based on three key methodological assumptions: similarity, consistency, and transitivity. Transitivity served as the theoretical basis for valid indirect comparisons [27]. Statistical heterogeneity was assessed using posterior estimates of the between-study variance (τ) from the Bayesian NMA model and summarized as I^2 statistics [28]. In the absence of closed loops in the NMA, inconsistency was assessed indirectly by comparing the results of the PWMA with those of the NMA, as formal inconsistency assessment is impossible without closed loops [29, 30]. Meta-regression analysis was not conducted, as the limited number of studies for several interventions precluded reliable estimation [31].

Publication bias was evaluated using Egger's regression test, with a p-value < 0.10 considered indicative of significant small-study effects. In addition, funnel plots were constructed to visually assess asymmetry. All statistical tests were two-sided, and a p-value < 0.05 was considered statistically significant. All analyses were performed using R software (version 4.4.3).

Results

Study characteristics and network plots

In comprehensive database searches conducted up to June 30, 2025, 1,354 records were identified, and an additional 224 records were retrieved through citation searching (Fig. 1). Records from all sources were merged before deduplication. After removal of 494 duplicate records, 1,084 records underwent title and abstract screening, of which 961 were excluded. Subsequently, 123 reports were sought for retrieval, of which 19 could not be retrieved. A total of 104

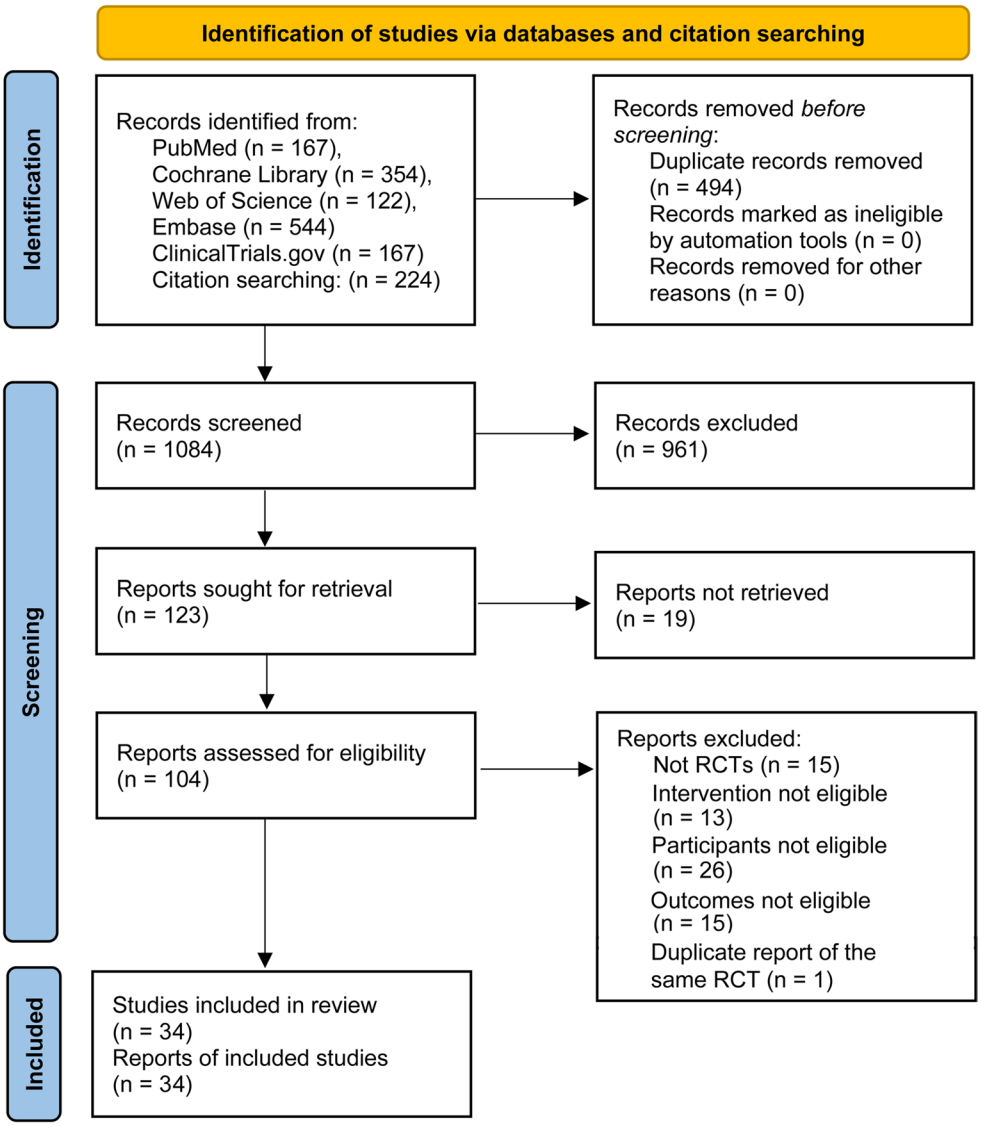


Fig. 1 PRISMA 2020 flow diagram of the study selection process for the NMA. PRISMA flow diagram showing the study selection process, including identification, screening, retrieval, eligibility assessment, and study inclusion. Records were identified through database searches and citation searching, merged before deduplication, and subsequently screened for eligibility. A total of 34 studies, comprising 35 RCTs, were included in the NMA

reports were assessed for eligibility, and 70 were excluded for predefined reasons. Ultimately, 34 studies, comprising 35 RCTs published between 2012 and 2025, were included in the NMA. One study contributed two independent RCTs. The study-specific diagnostic criteria, baseline characteristics and intervention details are summarized in Table S3. The numbers of studies and total participants contributing to each direct comparison by outcome are summarized in Table S4. Among the PBM studies, wavelengths ranged from 660 to 1070 nm, session durations from 10 to 20 min, and treatment courses from 2 to 16 weeks. The network plots for all outcomes (Fig. 2) each comprised four interventions: PBM, tDCS, rTMS, and a control group (CON). All networks were fully connected through CON, enabling indirect comparisons among active interventions.

Cognitive outcomes

The comparative effects of PBM, rTMS, and tDCS were evaluated based on changes from baseline in two global cognitive measures, MMSE and ADAS-Cog. The corresponding results are presented in Fig. 3A and B.

For MMSE, PBM, tDCS, and rTMS were all associated with cognitive improvements compared with CON. Among them, PBM demonstrated the highest estimated treatment effect (MD = 3.14, 95% CrI: 1.71 to 4.60), followed by tDCS (MD = 1.93, 95% CrI: 1.11 to 2.79) and rTMS (MD = 1.58, 95% CrI: 0.87 to 2.26). In indirect comparisons, no statistically significant differences were observed between PBM and rTMS (MD = 1.57, 95% CrI: −0.02 to 3.18), PBM and tDCS (MD = 1.21, 95% CrI: −0.48 to 2.87), or rTMS and tDCS (MD = −0.35, 95% CrI: −1.44 to 0.68).

For ADAS-Cog, in which lower scores indicate better cognitive performance, PBM, tDCS, and rTMS were all associated with cognitive improvements compared with CON. Among them, PBM demonstrated the greatest estimated improvement (MD = −8.01, 95% CrI: −12.82 to −3.13), followed by tDCS (MD = −3.30, 95% CrI: −6.10 to −0.52) and rTMS (MD = −2.77, 95% CrI: −4.16 to −1.46). In indirect comparisons, PBM was significantly more effective than rTMS (MD = −5.24, 95% CrI: −10.23 to −0.12), whereas no statistically significant differences were observed between PBM and tDCS (MD = −4.71, 95% CrI: −10.24 to 0.95) or between rTMS and tDCS (MD = 0.53, 95% CrI: −2.49 to 3.49).

Feasibility and safety

Dropout rates were used as an indicator of feasibility (Fig. 3C). Compared with CON, PBM (RR = 0.54, 95% CrI: 0.19 to 1.40), rTMS (RR = 0.72, 95% CrI: 0.44 to 1.01), and tDCS (RR = 0.62, 95% CrI: 0.21 to 1.64) showed lower estimated dropout risks, but none of these comparisons reached statistical significance. Comparisons among the

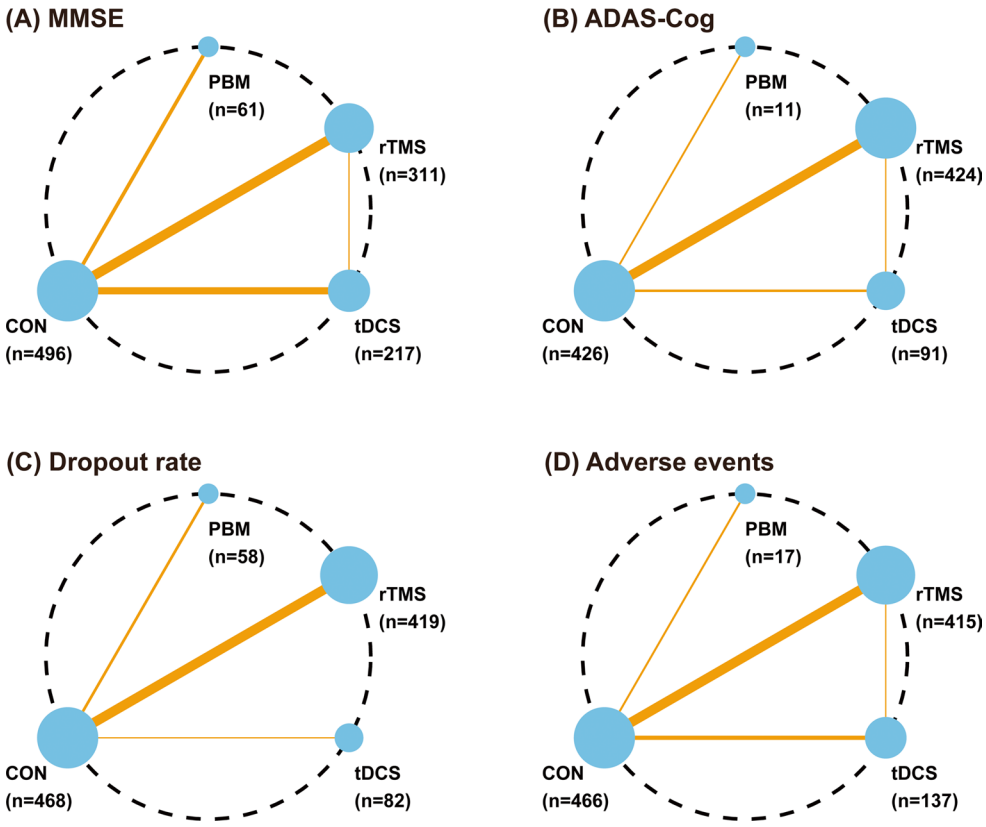


Fig. 2 Network plots of the Bayesian network meta-analysis for all outcomes. (**A**) MMSE; (**B**) ADAS-Cog; (**C**) dropout rate; (**D**) adverse events. Each node represents an intervention, with the node size proportional to the total number of participants assigned to that intervention. Connecting lines indicate direct comparisons between interventions, with line thickness proportional to the number of contributing trials. Abbreviations: MMSE, Mini-Mental State Examination; ADAS-Cog, Alzheimer's Disease Assessment Scale–Cognitive Subscale; PBM, photobiomodulation; rTMS, repetitive transcranial magnetic stimulation; tDCS, transcranial direct current stimulation; CON, control

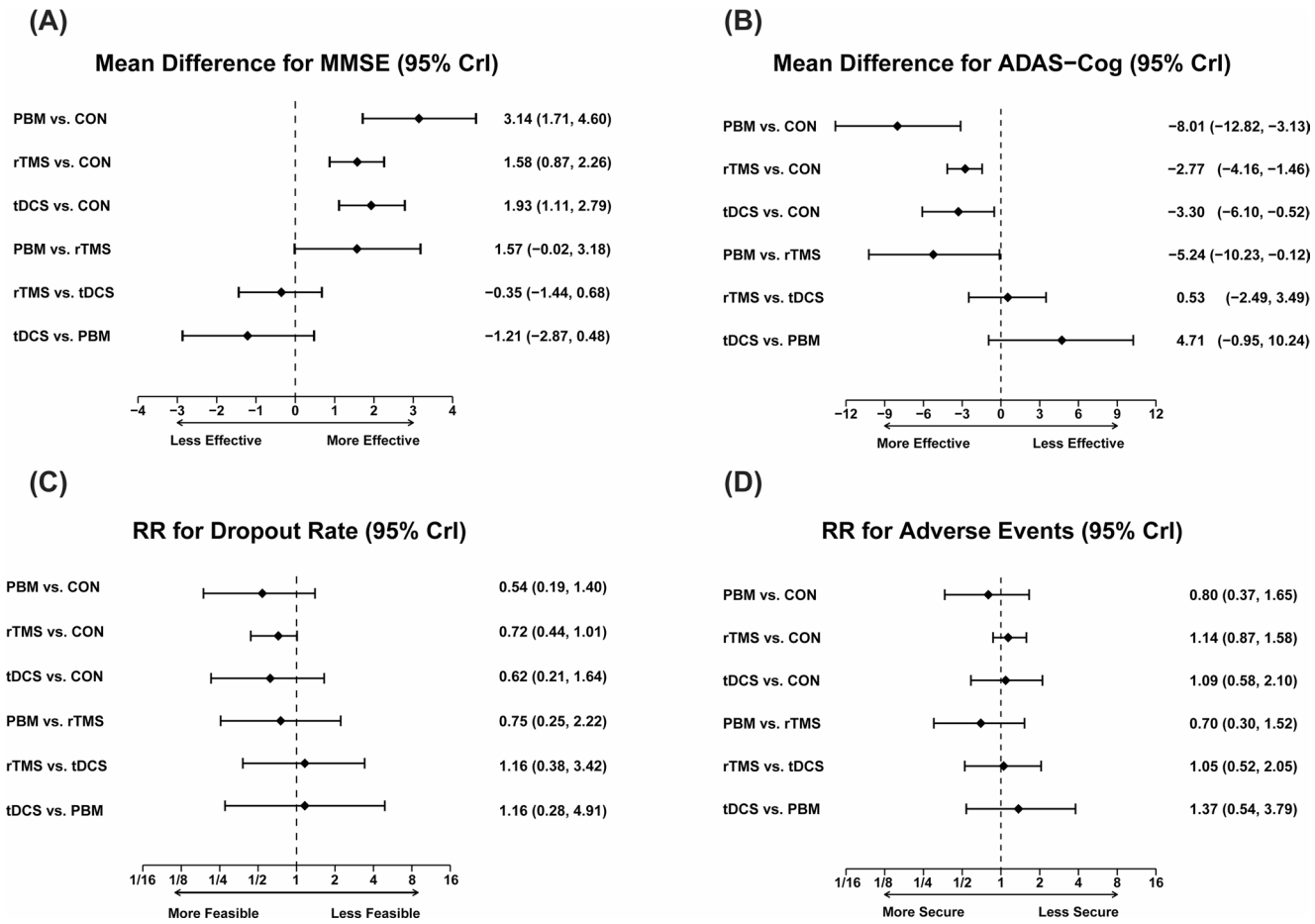


Fig. 3 Forest plots from the Bayesian network meta-analysis. (**A**) For MMSE, a mean difference (MD) greater than 0 indicates superior cognitive improvement of the specified intervention compared to the comparator (**B**) For ADAS-Cog, an MD less than 0 indicates superior cognitive improvement (**C**) For dropout rate, a risk ratio (RR) less than 1 indicates better feasibility (**D**) For adverse events, a RR less than 1 indicates greater safety. Effect sizes are presented with 95% credible interval (CrI). Abbreviations: MD, mean difference; RR, risk ratio; CrI, credible interval

interventions did not reveal statistically significant differences: PBM compared with rTMS (RR = 0.75, 95% CrI: 0.25 to 2.22), rTMS compared with tDCS (RR = 1.16, 95% CrI: 0.38 to 3.42), and tDCS compared with PBM (RR = 1.16, 95% CrI: 0.28 to 4.91).

Safety was evaluated based on the incidence of adverse events (Fig. 3D). Compared with CON, none of the interventions was associated with a statistically significant increase in adverse event risk: PBM (RR = 0.80, 95% CrI: 0.37 to 1.65), rTMS (RR = 1.14, 95% CrI: 0.87 to 1.58), and tDCS (RR = 1.09, 95% CrI: 0.58 to 2.10). Similarly, no significant differences were observed among the interventions: PBM compared with rTMS (RR = 0.70, 95% CrI: 0.30 to 1.52), rTMS compared with tDCS (RR = 1.05, 95% CrI: 0.52 to 2.05), and tDCS compared with PBM (RR = 1.37, 95% CrI: 0.54 to 3.79). Overall, these secondary outcomes should be interpreted as exploratory and descriptive.

Rankings

The rankings were estimated using Bayesian models based on the posterior distribution of possible ranks across interventions (Fig. 4; Table S5). Higher ranks correspond to greater improvement in cognitive outcomes, lower dropout risk, and fewer adverse events, according to the outcome assessed. SUCRA values provided a quantitative synthesis of ranking probabilities across interventions. For cognitive outcomes, PBM ranked first in both MMSE (probability of ranking first = 92.01%, SUCRA = 0.97) and ADAS-Cog (94.62%, SUCRA = 0.98). tDCS and rTMS ranked below PBM, with tDCS (SUCRA: 0.61 for MMSE; 0.56 for ADAS-Cog) ranking above rTMS (SUCRA: 0.43 for MMSE; 0.46 for ADAS-Cog). For feasibility, PBM again ranked first (51.37%; SUCRA = 0.73), followed by tDCS (SUCRA = 0.62) and then rTMS (SUCRA = 0.55). Similarly,

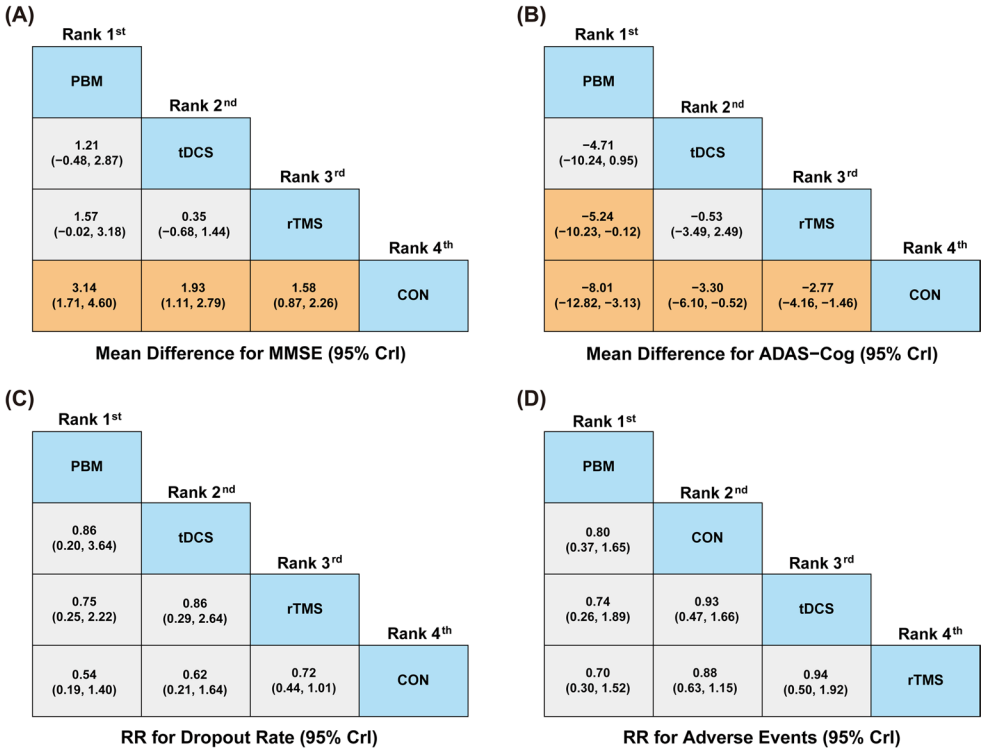


Fig. 4 Relative treatment rankings from the Bayesian network meta-analysis. (**A**) MDs and 95% CrI for MMSE, and an MD greater than 0 indicates superior cognitive improvement of the column-defined intervention compared to the row-defined intervention (**B**) MDs and 95% CrI for ADAS-Cog, and an MD less than 0 indicates superior cognitive improvement (**C**) RRs and 95% CrI for dropout rate, and a RR less than 1 indicates better feasibility (**D**) RRs and 95% CrI for adverse events, and a RR less than 1 indicates greater safety. Orange-shaded cells represent statistically significant differences, while grey cells indicate non-significant results

PBM ranked first for safety (64.46%; SUCRA = 0.77), and tDCS (SUCRA = 0.39) was again ranked above rTMS (SUCRA = 0.26) for this outcome. Overall, PBM ranked first across all outcomes. CON was ranked last in cognitive and feasibility outcomes, and second in safety. Although PBM ranked above CON for safety, this ordering should not be interpreted as evidence of a statistically significant safety advantage of PBM over CON. These rankings provide a descriptive summary of the relative ordering of interventions and should be interpreted alongside the corresponding effect estimates and credible intervals.

Subgroup analysis

To explore sources of potential heterogeneity, subgroup analyses were conducted. Studies were classified according to mean baseline MMSE (> 20 vs. ≤20) and intervention duration (> 4 weeks vs. ≤4 weeks), following established stratification approaches [32–34]. Due to the limited availability of other outcomes, subgroup analysis focused solely on MMSE (Fig. 5). Across all subgroups, active interventions showed cognitive benefits compared with CON, with PBM consistently achieving the highest ranking, followed by tDCS and rTMS. This ranking was identical to the primary analysis, and no significant differences were observed among active interventions, except for PBM showing a statistically significant advantage over rTMS in the mean baseline MMSE > 20 subgroup. Detailed effect sizes, pairwise estimates, and heterogeneity statistics are provided in Table S6.

Risk of bias, publication bias, inconsistency, and heterogeneity

Risk of bias was assessed using the Cochrane RoB 2 tool. Among the included RCTs, 71.4% (25/35), 25.7% (9/35), and 2.9% (1/35) were judged to have an overall low risk, some concerns, and high risk of bias. Most studies had a low risk of bias in the domains of randomization (32/35, 91.4%), missing outcome data (35/35, 100%), and selective reporting (35/35, 100%). The primary risk of bias arose in the domains of deviations from intended interventions (4/35, 11.4%) and outcome measurement (5/35, 14.3%). Detailed domain-level assessments are presented in Fig. S1 and Fig. S2.

Publication bias was assessed using funnel plots and Egger's regression tests. At the network level, the comparison-adjusted funnel plot showed no substantial asymmetry, suggesting an absence of notable publication bias (Fig. S3). Consistently, the Egger meta-regression of centered mean differences on study SE, with treatment comparison as a fixed effect, yielded a non-significant slope for MMSE ($\beta = 0.11$; $p = 0.84$), indicating no evidence of small-study effects. For ADAS-Cog, dropout, and adverse events, the number of studies per comparison was insufficient to permit Egger's regression tests. Corresponding funnel plots revealed no substantial asymmetry.

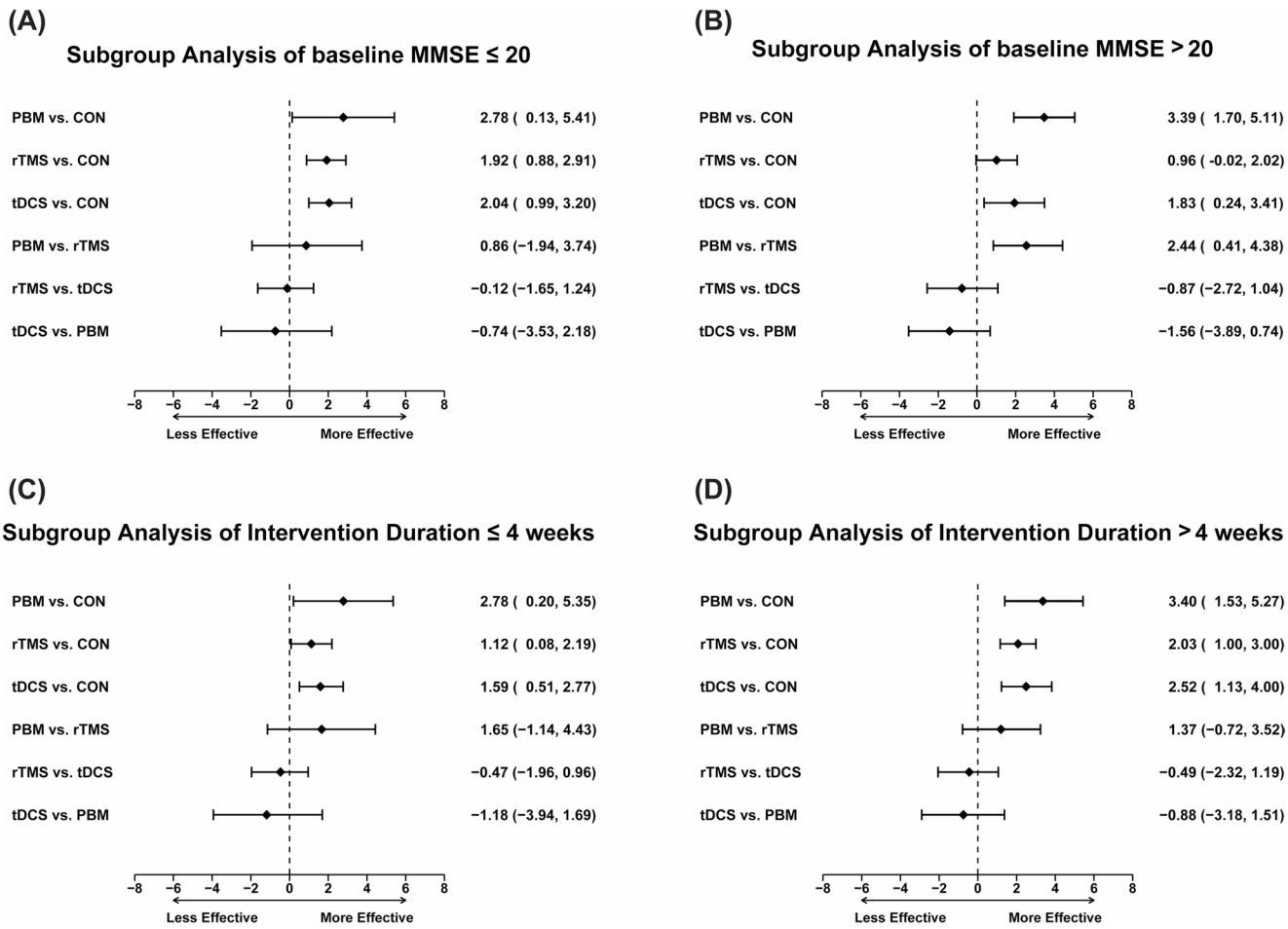


Fig. 5 Subgroup analyses of MMSE from the Bayesian network meta-analysis (**A**, **B**) Subgroups based on baseline MMSE scores ≤ 20 and > 20, respectively (**C**, **D**) Subgroups based on intervention duration ≤ 4 weeks and > 4 weeks, respectively. Effect estimates are presented as MDs with 95% CrI; an MD greater than 0 indicates superior cognitive improvement of the specified intervention compared to the comparator

Formal inconsistency assessment was not feasible because no closed loops were present in the evidence networks. As an alternative, we assessed it indirectly by comparing effect estimates from PWMA with those generated by NMA. The similarity in effect sizes supported the assumption of network consistency (Table S7).

Heterogeneity was evaluated using the posterior estimates of between-study variance (τ) and summarized as I^2 statistics for each outcome. In the primary analysis, the degree of heterogeneity varied across outcomes, with I^2 values of 46.5% for MMSE, 62.1% for ADAS-Cog, 8.5% for dropout rate, and 15.7% for adverse events, indicating low to moderate heterogeneity that was considered acceptable for this NMA (Table S7). Notably, heterogeneity was lower in the subgroup with baseline MMSE > 20, whereas subgroup analyses by intervention duration did not show a clear reduction in heterogeneity. These findings suggest that baseline cognitive level may partially explain the observed between-study heterogeneity, thereby improving the stability and interpretability of the NMA model.

In addition, the convergence of the NMA model was assessed using trace plots (Fig. S4), and the Brooks–Gelman–Rubin diagnostic indicated the stability and reliability of posterior sampling across MCMC chains (Fig. S5).

Discussion

Our study represents the first NMA to assess the cognitive efficacy, feasibility, and safety of PBM, tDCS, and rTMS in patients with AD. Based on data from 35 RCTs, our analysis showed that all three NIBS modalities were associated with improvements in global cognitive outcomes compared with CON. PBM yielded the largest estimated effect versus CON on both MMSE and ADAS-Cog; tDCS and rTMS also showed statistically significant benefits versus CON, although their effect sizes were relatively modest. In between-intervention comparisons, PBM showed a significant advantage over rTMS for ADAS-Cog, whereas no statistically significant difference was observed for MMSE. The comparisons

between PBM and tDCS were not statistically significant for either cognitive outcome. As the evidence network was primarily connected through CON, these between-intervention comparisons should be interpreted as comparative estimates derived largely from indirect evidence.

In outcome-specific SUCRA-based rankings, PBM ranked highest for both cognitive outcomes. This pattern was broadly consistent with the NMA estimates: PBM showed a statistically significant advantage over rTMS for ADAS-Cog, whereas the corresponding comparison for MMSE and the comparisons with tDCS were not statistically significant. Moreover, although most included RCTs were judged to have a low overall risk of bias, a subset of studies raised concerns, chiefly regarding deviations from intended interventions and outcome measurement. Taken together, these considerations suggest that the SUCRA hierarchy is more informative for the overall ordering of interventions within each outcome than for fine distinctions between active treatments, particularly when between-intervention differences are statistically non-significant. For the secondary outcomes, none of the interventions increased the risk of dropout or adverse events relative to CON. Although PBM ranked first by SUCRA for feasibility and safety, the differences across interventions were not statistically significant, supporting a cautious interpretation of these secondary outcomes. Subgroup analyses stratified by baseline MMSE and intervention duration were directionally consistent with the primary analysis, supporting the robustness of the results across the examined subgroups.

Previous meta-analyses have evaluated the cognitive effects of NIBS in patients with AD, with most focusing on tDCS and rTMS. Many reported modest to moderate improvements in cognitive function versus CON, consistent with the direction of our findings. For example, Šimko et al. and Majdi et al. reported significant cognitive benefits with rTMS and tDCS, respectively [35, 36]. However, Su et al. described no significant overall improvement with tDCS [37]. Beyond differences in reported efficacy, several reviews grouped different NIBS modalities into a single category, without considering distinct mechanisms or clinical applicability [17]. In addition, new modalities such as PBM were often excluded due to limited clinical evidence [13]. Even when PBM was included, it was frequently classified as light therapy rather than as NIBS [18]. These practices obscured intervention-specific effects and reduced interpretability. In particular, classifying PBM as light therapy limited its inclusion in comparative analyses with other NIBS. In our study, PBM was classified as a NIBS intervention because it delivers non-invasive transcranial photic stimulation with red or near-infrared light, similar in approach to tDCS and rTMS [38, 39]. We evaluated PBM, tDCS, and rTMS as distinct interventions within a Bayesian NMA that synthesized direct comparisons versus CON and yielded indirect comparisons among active interventions. SUCRA was used to summarize ranking probabilities across interventions. This framework avoided pooling all NIBS into a single category and facilitated clinically interpretable intervention-specific comparisons.

Another limitation of prior meta-analyses was their focus on cognitive outcomes, with limited consideration of feasibility and safety [40]. Such a restrictive outcome framework may overestimate clinical applicability and overlook implementation challenges. In particular, interventions that require high procedural precision, specialized equipment, or continuous supervision may be impractical for older adults or individuals with functional impairments, even when cognitive effects are favorable. To improve clinical interpretability, we evaluated three outcome domains: cognitive efficacy, feasibility (dropout), and safety (adverse events). In SUCRA summaries, PBM often ranked highest across cognition, feasibility, and safety. These ranks are descriptive and complement the NMA estimates, indicating that cognitive outcomes alone may not capture the full clinical value.

To maintain valid and interpretable comparisons across trials, it was essential to reduce heterogeneity in populations and intervention protocols. However, some prior studies enrolled heterogeneous populations by including individuals with AD, MCI, or even non-AD dementias [41, 42]; others combined NIBS interventions with adjunctive treatments such as cognitive behavioral therapy, exercise programs, or pharmacological agents [43]; and several studies lacked subgroup analyses to account for variations in baseline cognitive level or intervention duration [44]. We addressed these concerns by implementing several methodological strategies. First, we restricted inclusion to studies enrolling participants with AD or probable AD, with the study-specific diagnostic criteria summarized in Table S3. Second, we excluded trials that combined PBM, tDCS, or rTMS with other interventions. Third, we conducted subgroup analyses based on mean baseline MMSE scores and intervention duration to assess whether the treatment effects remained consistent across clinical subgroups. Collectively, these strategies improved comparability across trials and supported the interpretability of the network estimates.

The cognitive efficacy of PBM observed in our NMA may be attributed to its distinctive neurobiological mechanisms. tDCS and rTMS primarily act by modulating cortical excitability and synaptic plasticity. tDCS modulates neuronal excitability through current-induced shifts in membrane potential [45, 46], whereas rTMS promotes LTP-like synaptic plasticity via frequency-dependent magnetic stimulation [47, 48]. These effects may facilitate reorganization and functional network integration within large-scale cognitive networks, which are known to be disrupted in AD [49–51]. In contrast, PBM more directly influences cellular bioenergetics

by targeting mitochondrial cytochrome c oxidase [52–54], thereby enhancing oxidative phosphorylation and supporting synaptic energy metabolism. In a neurodegenerative context such as AD, where mitochondrial dysfunction and impaired energy metabolism are closely linked to disease pathophysiology, this more proximal bioenergetic mechanism may partly explain the more favorable ranking of PBM in our analysis, although current mechanistic insights are largely derived from animal or cellular models [55]. The PBM evidence contributing to the present analysis was derived from red or near-infrared protocols using wavelengths of 660–1070 nm, session durations of 10–20 min, and treatment courses of 2–16 weeks; however, target configurations, power, and pulse frequency varied or were not consistently reported, precluding identification of a single dominant protocol.

Beyond these mechanistic considerations, PBM also demonstrated practical clinical applicability, particularly among older adults. These individuals frequently experience frailty, multiple comorbidities, and limited caregiving support [56, 57]. Under such conditions, many NIBS interventions may face implementation challenges related to procedural complexity and patient tolerability. For example, tDCS often requires precise electrode placement, continuous supervision, and fine-tuning of stimulation parameters, which may be difficult to apply effectively in frail or cognitively impaired individuals [58]. Similarly, rTMS typically requires bulky equipment, precise coil placement, and multiple sessions delivered in specialized settings, which can make implementation challenging for patients living in community or home settings [59]. In contrast, PBM devices generally deliver low-level red or near-infrared light through simple LED arrays and are developed as lightweight, helmet-like wearables [60, 61]. Such devices can often be applied in a non-contact mode, with simplified procedures and minimal reliance on specialized equipment. Given these features, PBM may be feasibly implemented in long-term care facilities, community-based programs, and even home settings, where structured clinical resources may be lacking. Taken together, PBM combines plausible mechanistic relevance with practical features that may support adherence and implementation in AD, although these factors were not formally included as comparative variables in the present NMA.

This study had some limitations that should be acknowledged. First, although most included RCTs used sham controls, the effectiveness of blinding may still have been compromised by the technical limitations of commercial devices. In NIBS trials, perceptible sensations such as tingling or scalp discomfort and visible device features can undermine blinding integrity, potentially introducing performance or detection bias [62]. Second, the clinical evidence for PBM in AD remains preliminary [63]. Because direct head-to-head evidence among active interventions was very limited, particularly for PBM versus tDCS or rTMS, between-intervention comparisons involving PBM relied largely on indirect evidence. Transitivity also could not be fully verified given differences in stimulation parameters, target sites, treatment schedules, settings, and disease severity, warranting caution in interpreting relative rankings. Third, although studies explicitly enrolling non-AD dementias were excluded and diagnostic information was summarized in Table S3, formal diagnostic criteria were not consistently reported across all trials, which may introduce residual diagnostic uncertainty. Finally, the small evidence base precluded meta-regression analyses, limiting exploration of how intervention parameters, treatment duration, target site, session schedule, disease severity, and outcome-monitoring windows might modify treatment effects.

Future research should be supported by a larger pool of high-quality RCTs. In particular, head-to-head RCTs directly comparing PBM with tDCS or rTMS are needed to reduce reliance on indirect comparisons. Future studies should also incorporate domain-specific cognitive outcomes to more comprehensively evaluate intervention effects. In addition, to clarify how stimulation parameters such as intensity, frequency, and duration influence efficacy, future trials should adopt standardized protocols, prespecified stratification, and sufficient sample sizes to enable meaningful subgroup or dose–response analyses and to inform personalized intervention strategies. As the randomized evidence base expands, future comparative syntheses may also incorporate emerging NIBS modalities in AD.

Conclusion

This Bayesian NMA indicated that, within the current evidence network, PBM showed the highest probability of ranking first for both global cognitive outcomes in patients with AD, as measured by MMSE and ADAS-Cog. Nevertheless, the available indirect comparisons did not establish consistent evidence of a statistically significant advantage of PBM over rTMS and tDCS across MMSE and ADAS-Cog. Beyond efficacy, feasibility and safety were broadly comparable across interventions. Considering practical factors such as simpler setup and potential for home-based delivery, PBM may be a feasible option for some older adults with frailty or multimorbidity. These implementation aspects were not prespecified endpoints and were not formally compared in this analysis. Our findings highlight PBM as a promising NIBS approach for AD and provide a clear rationale for future head-to-head RCTs with standardized protocols, longer follow-up periods, and domain-specific cognitive outcomes to clarify comparative efficacy and clinical applicability.

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Author contributions Author Contributions StatementX.W. and K.Z. conceived and designed the study. X.W. performed the data curation and formal analysis, developed the methodology and software, prepared the figures, and wrote the main manuscript text.J.C. contributed to data curation and formal analysis. Y.Z. contributed to data curation, validation, and resources. C.L. contributed to formal analysis, methodology, and software. Y.H. contributed to the methodology and software. K.Z. supervised the study, acquired funding, administered the project, and critically revised the manuscript. All authors read and approved the final manuscript.

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Data availability The aggregated data and analysis code underlying this network meta-analysis are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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